

Q. Write a bash script to create a menu driven program to do the following:

- A welcome message with your name and UID
- Add menu:
 1. Add User
 2. List number of Items
 3. Check Permission
 4. List Processes
 5. Exit

NOTE: User should be able to choose any of the given options to perform the corresponding task.

Q. Create a Command Shell Environment on machine. :

- Create a command/redirection/alias named as 'qstat'.
- It should be able to execute the following command:
`ps -eo pid,tid,class,rtprio,ni,pri,psr,pcpu,stat,wchan,comm`

Q. Deny cron access to user mike (Create if not available). It should not affect other users.

Q. Find files which are owned by user "natasha" (Create if not available) and copy them to "/natasha_secret"

Q. Regular Expressions:

- a) Use grep with a regular expression to find all lines in /tmp/regex_test/system.log that contain IP addresses (both IPv4 and IPv6 formats).
- b) Use grep to find all lines that contain either "Error" or "Warning" (case-insensitive).
- c) Use grep to find all lines that end with a number.

Setup Commands (Run before attempting this section):

```
# mkdir -p /tmp/regex_test
# echo "Error: Failed to connect to database at 10.0.0.1" > /tmp/regex_test/system.log
# echo "Warning: Disk space low on /dev/sda1" >> /tmp/regex_test/system.log
# echo "Info: User admin logged in from 192.168.1.50" >> /tmp/regex_test/system.log
# echo "Error: Authentication failed for user guest" >> /tmp/regex_test/system.log
# echo "Info: Backup completed successfully" >> /tmp/regex_test/system.log
```

Q. Web Server Log Analysis, Analyze Apache web server logs to extract insights about traffic, errors, and user behavior. (20 Marks)

[+] Objectives:

- Parse web server log files
- Extract specific information using grep and awk
- Generate statistics from log data
- Create summary reports

1. Count the total number of requests in the /tmp/weblogs/access.log - (5 Marks)
2. Find Error Requests: Find all requests that returned error status codes (4xx or 5xx) - (5 Marks)
3. Extract Unique IP Addresses: Extract and count unique IP addresses from the log - (5 Marks)
4. Find Most Requested Page: Identify the most frequently requested page (Extract the request path, count occurrences, and find the most frequent) - (5 Marks)

Setup Commands (Run before attempting this section): to make sample access.log:

```
# mkdir -p /tmp/weblogs
# echo '192.168.1.100 - - [10/Oct/2023:13:55:36 +0000] "GET /index.html HTTP/1.1" 200 2326 "-" "Mozilla/5.0"' > /tmp/weblogs/access.log
# echo '192.168.1.101 - - [10/Oct/2023:13:56:12 +0000] "GET /about.html HTTP/1.1" 200 1547 "http://example.com/" "Mozilla/5.0"' >> /tmp/weblogs/access.log
# echo '192.168.1.102 - - [10/Oct/2023:13:57:23 +0000] "POST /login HTTP/1.1" 401 89 "-"
```

```
"curl/7.68.0" >> /tmp/weblogs/access.log
# echo '192.168.1.100 - - [10/Oct/2023:13:58:45 +0000] "GET /images/logo.png HTTP/1.1" 200 15324
"http://example.com/index.html" "Mozilla/5.0"' >> /tmp/weblogs/access.log
# echo '192.168.1.103 - - [10/Oct/2023:13:59:01 +0000] "GET /admin HTTP/1.1" 403 112 "-" "python-
requests/2.25.1"' >> /tmp/weblogs/access.log
```

Q. Create a system monitoring dashboard (shell script) by collecting and analyzing various system metrics.

[+] Objectives:

- Collect CPU and memory usage data
- Monitor running processes
- Track disk space usage
- Generate system health reports

1. Check CPU Usage: Check current CPU usage and load average [Hint: top -bn1 | head -5]
2. Monitor Memory Usage: Check memory usage and available memory [Hint: free -h]
3. List Top Processes: List top 10 processes by CPU usage [Hint: ps aux --sort=-%cpu | head -11]